

TRIASHA SARKAR

APPLIED ML ENGINEER | ML/AI ENGINEER | SCIENTIFIC ML ENGINEER

Atlanta, GA | +1 (404) 948-8761 | tsarkar34@gatech.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Applied ML/AI engineer and Georgia Tech M.S. candidate with 1.5 years of commercial systems-engineering experience. Builds production-oriented ML systems spanning RAG/LLM applications, recommendation and search, scientific ML, uncertainty quantification, and optimization, with rigorous evaluation, FastAPI/Docker/cloud deployment, CI/CD, and observability.

TECHNICAL SKILLS

Languages and data: Python, SQL, C++ (working), MATLAB; pandas, NumPy, SciPy, DuckDB

ML and GenAI: scikit-learn, PyTorch, TensorFlow; classification, regression, recommender systems, RAG, BM25, dense retrieval, Sentence Transformers, RRF, cross-encoder reranking, grounded generation

Scientific ML and evaluation: surrogate modeling, uncertainty quantification, conformal prediction, constrained multi-objective optimization, physics validation; held-out evaluation, NDCG/MRR/Recall, ROC AUC, bootstrap confidence intervals

ML systems and cloud: FastAPI, REST APIs, Docker/Compose, Google Cloud Run, Cloud Storage, ONNX Runtime, Prometheus, OpenTelemetry, GitHub Actions, pytest, Ruff, mypy, CI/CD

EXPERIENCE

Graduate Research Assistant | Aerospace Systems Design Laboratory, Georgia Tech

2025-2026

- Built and validated Python/scikit-learn workflows for simulation reliability and uncertainty-aware decision support, covering data audit, preprocessing, feature engineering, model evaluation, reproducibility, and technical communication.
- Audited a **15,120-case** simulation campaign and found **6,804 silent failures**; removed duplicate-geometry leakage and trained a classifier with **96.8% accuracy** and **0.986 ROC AUC** on unseen geometries, then replicated results on an independent 1,050-case study.
- Co-developed GREEN TEA techno-economic and well-to-wake lifecycle models; benchmarked outputs against GREET, CORSIA, and 40BSAF-GREET and delivered a documented **model now used by the sponsor**.

Systems Engineer | Alten India, embedded at Rolls-Royce

Oct 2023-Apr 2025

Bangalore, India

- Validated engineering models supporting Trent XWB-84 EP airworthiness certification, maintaining traceable evidence and communicating technical findings across multidisciplinary teams in a regulated commercial environment.

SELECTED AI/ML AND SCIENTIFIC ML PROJECTS

AeroRAG-X: Traceable Retrieval for Aerospace Technical Knowledge

2026

Independent project | github.com/triasha72/AeroRAG-X

- Built an **end-to-end evidence-grounded RAG system** over **3,233 NASA NTRS chunks** using BM25 + Sentence Transformer dense retrieval, Reciprocal Rank Fusion, cross-encoder reranking, facet-aware evidence selection, sufficiency gating, OpenAI Responses API generation, and claim-level citations.
- Reached **100% answerability accuracy, unsupported-refusal, claim-citation coverage, and citation-reference validity** on a 32-query development benchmark; added a separate held-out evaluation and CI regression policy, then deployed a private FastAPI/Docker service to **Google Cloud Run** with Prometheus metrics, OpenTelemetry tracing, and P50/P95/P99 load validation.

EdgeGenBench: Uncertainty-Aware Surrogates, Optimization, and Edge AI

2026

Independent project | github.com/triasha72/EdgeGenBench

- Built a **reproducible scientific-ML benchmark** for hybrid-electric and hydrogen aircraft design using synthetic physics data, Ridge/Random Forest/HistGradientBoosting surrogates, conformal and tree-quantile uncertainty, false-safe-rate-aware classification, constrained multi-objective optimization, and physics validation.
- Achieved **0.995 mean test R^2** and **99.89% classifier ROC AUC** with a **2.12% false-safe rate**; obtained **100% physics-feasibility agreement** on optimizer-selected designs and exported models to ONNX, reducing batch-1 inference from **13.8-14.8 ms to 0.305-0.311 ms**.

NewsLens: Leakage-Aware News Recommendation, Search, and Serving

2026

Independent project | github.com/triasha72/NewsLens

- Built a **leakage-aware recommendation pipeline** over **5.84M candidate interactions** using Python/SQL/DuckDB, chronological splitting, TF-IDF personalization, and a training-only popularity fallback; achieved **NDCG@10 0.3664, Recall@10 0.5955, and Hit Rate@10 0.6762**.
- Packaged versioned model artifacts behind FastAPI** with readiness, request tracing, and latency reporting; added bootstrap confidence intervals and subgroup/failure analysis, containerized with non-root Docker/Compose, published multi-architecture images via GitHub Actions, and maintained **400+ automated tests**.

EDUCATION

M.S., Aerospace Engineering | Georgia Institute of Technology

Aug 2026

Atlanta, GA

M.Sc., Aerospace Vehicle Design | Cranfield University

Feb 2023

United Kingdom | First Class Honours

B.Tech., Aerospace Engineering | SRM Institute of Science and Technology

Jun 2021

India | First Class Honours